

Practical 2

1. Aim: Write a shell script to generate mark- sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student.

```
HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ #!/bin/bash

echo "Enter marks of Subject 1:"
read m1

echo "Enter marks of Subject 2:"
read m2

echo "Enter marks of Subject 3:"
read m3

total=$((m1 + m2 + m3))
percentage=$((total / 3))

echo "Total Marks = $total"
echo "Percentage = $percentage%"

if [ $percentage -ge 60 ]
then
    echo "Class: First Class"
elif [ $percentage -ge 50 ]
then
    echo "Class: Second Class"
elif [ $percentage -ge 40 ]
then
    echo "Class: Pass"
else
    echo "Class: Fail"
fi

Enter marks of Subject 1:
78
Enter marks of Subject 2:
34
Enter marks of Subject 3:
77
Total Marks = 189
Percentage = 63%
Class: First Class

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ |
```

2. Aim: Write a menu driven shell script which will print the following menu and execute the given task.

- 1) Display calendar of current month.

```
HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ $ #!/bin/bash
date +"%B %Y"
bash: $: command not found
January 2026
```

- 2) Display today,s date and time.

```
HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ date
Sat Jan 24 22:57:32 IST 2026
```

3) Display usernames those are currently online

```
HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ whoami
HP
```

4) Display your terminal number.

```
HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ echo $TERM
xterm
```

3. Aim: Write a shell script which will generate first n Fibonacci numbers like: 1, 1, 2, 3, 5, 13

```
HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ #!/bin/bash

echo "Enter number of terms:"
read n

a=1
b=1

echo "Fibonacci Series:"
echo -n "$a $b "

for ((i=3; i<=n; i++))
do
    c=$((a + b))
    echo -n "$c "
    a=$b
    b=$c
done
echo
Enter number of terms:
4
Fibonacci Series:
1 1 2 3
```

4. Write a shell script which will accept a number b and display first n prime numbers as output.

```

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ #!/bin/bash

echo "Enter how many prime numbers you want:"
read n

count=0
num=2

echo "Prime numbers are:"

while [ $count -lt $n ]
do
    flag=1
    for ((i=2; i<=num/2; i++))
    do
        if [ $((num % i)) -eq 0 ]
        then
            flag=0
            break
        fi
    done

    if [ $flag -eq 1 ]
    then
        echo -n "$num "
        count=$((count + 1))
    fi

    num=$((num + 1))
done
Enter how many prime numbers you want:
7
Prime numbers are:
2 3 5 7 11 13 17

```

5. Aim: Write menu driven program for file handling activity.

1) Creation of file.

```

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ echo "Enter file name:"
read fname
touch
"$fname"
echo "File created"
Enter file name:
ayushi.txt
touch: missing file operand
Try 'touch --help' for more information.
bash: ayushi.txt: command not found
File created

```

2) Write content in the file.

```

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ #!/bin/bash

echo "Enter file name:"
read fname

echo "Enter content (Press Ctrl + D to save):"
cat > $fname

echo "File saved successfully!"
Enter file name:
ayushi.txt
Enter content (Press Ctrl + D to save):
hey ayushi!
File saved successfully!

```

3) Append file content.

```
HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ #!/bin/bash

echo "Enter file name:"
read fname

echo "Enter content to append (Press Ctrl + D to save):"
cat >> $fname

echo "Content appended successfully!"
Enter file name:
ayushi.txt
Enter content to append (Press Ctrl + D to save):
hello
Content appended successfully!

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ |
```

4) Delete file content.

```
HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ #!/bin/bash

echo "Enter file name:"
read fname

> $fname

echo "File content deleted successfully!"
Enter file name:
ayushi.txt
File content deleted successfully!

HP@DESKTOP-B4BD73N MINGW64 ~/OneDrive/Desktop (main)
$ |
```